1. Recall is calculated as:

$$\text{Recall} = \frac{TP}{TP + FN}$$

From the confusion matrix:

- $TP = 6$
- $FN = 3$

Therefore,

$$\text{Recall} = \frac{6}{6+3} = \frac{6}{9} = 0.67$$

Correct answer: 0.67

2. Classifier A has **0% accuracy** → it makes no correct predictions.
   Classifier B has **50% accuracy** → it makes correct predictions half the time.
   Therefore, **B is more useful**.
3. A **larger** $k$is not always better in k-fold cross-validation.
   Larger k→ more training data per fold, but **higher computational cost** and potentially higher variance.
   Smaller k(e.g., **5 or 10**) often provides a good balance between computational efficiency and reliable performance estimation. So the answer is False.
4. For **20-fold cross-validation**:

$$\text{Training size} = \frac{19}{20} \times 1000 = 950$$

   Each fold contains $1000/20 = 50$samples for validation, so the remaining **950 samples** are used for training.
   Correct answer: 950
5.

5. Which of the following are true?

TP - True Positive, TN - True Negative, FP - False Positive, FN - False Negative

- [x] $\text{Precision} = \frac{TP}{TP+FP}$
- [x] $\text{Recall} = \frac{TP}{TP+FN}$
- [x] $\text{Accuracy} = \frac{2(TP+TN)}{TP+TN+FP+FN}$
- [ ] $\text{Recall} = \frac{FP}{TP+FP}$

6. False Positive rate and True Positive Rate.
7. A) **It helps reduce variance**
   B) **If the parameters of the resultant classifiers are fully correlated, then bagging is inefficient.**

Why?

Bagging (Bootstrap Aggregating) primarily reduces **variance** by combining predictions from multiple models.

If the classifiers are **highly correlated**, they make similar errors, so averaging them provides little benefit.

Bagging generally **does not reduce bias**.

8. **Correct answer: Stacking, because each classifier is trained on all of the available data**

   Reason: With limited training data, **stacking** is preferable because different base classifiers can learn from the **entire available dataset**, allowing you to exploit different learning algorithms without splitting the data among models.

9. Option 3 and 4